

Mathematical proof - Set 2 - Solutions

Specification Prf1-Prf4.

Answer key

Question	Answer	Topic
1	D	Mathematical proof
2	C	Mathematical proof
3	A	Mathematical proof
4	A	Mathematical proof
5	A	Mathematical proof
6	A	Mathematical proof
7	F	Mathematical proof
8	E	Mathematical proof
9	C	Mathematical proof

Worked solutions

Question 1 PRF-05 · Mathematical proof · spec Prf3

The sums $1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7$ are $1, 4, 9, 16$. A student conjectures that the sum of the first n odd numbers is n^2 . Which of the following is a valid justification?

Answer: D

A applies the arithmetic-series formula to all n . (C also gives a correct calculation but B, D and E are not justifications.)

Question 2 PRF-06 · Mathematical proof · spec Prf1

Which of the following statements can be disproved by a single counterexample?

- I Every prime number greater than 2 is odd.
- II Every integer of the form $4k + 3$ is prime.
- III There exists an even prime number.

Answer: C

Only a 'for all' statement can be disproved by one counterexample. I is true (no counterexample exists). II is false and $k = 3$ (15) disproves it. III is an existence statement - a counterexample is meaningless; it is proved by the example 2.

Question 3 PRF-09 · Mathematical proof · spec Prf1

Here is a proof that there are infinitely many primes, with the steps out of order.

- I Then N leaves remainder 1 on division by each of $p_1, \dots, p_n$, so no p_i divides N .
- II Suppose there are only finitely many primes $p_1, p_2, \dots, p_n$, and let $N = p_1 p_2 \cdots p_n + 1$.
- III So N has a prime factor that is not in the list, which is a contradiction.
- IV But $N > 1$, so N has at least one prime factor.

Which is the correct order?

Answer: A

Define N (II), show no listed prime divides it (I), note that N nevertheless has some prime factor (IV), and conclude that this factor is a new prime (III). The conclusion III cannot come before both I and IV have been established.

Question 4 PRF-10 · Mathematical proof · spec Prf1

Consider the claim: **if x and y are real numbers with $x + y > 10$, then $x > 5$ or $y > 5$.** Which of the following proves it?

Answer: A

A is proof by contrapositive/contradiction. B assumes the conclusion, C is an example, D proves something false, E's conclusion ('both exceed 5') is wrong.

Question 5 PRF-11 · Mathematical proof · spec Prf3

A student notices that $1 + 2 = 3$, $4 + 5 + 6 = 7 + 8$, $9 + 10 + 11 + 12 = 13 + 14 + 15$ and conjectures that the pattern continues. Which of the following describes the n th line of the pattern?

Answer: A

Line 1 starts at 1, line 2 at 4, line 3 at 9: start n^2 . Left sides have 2, 3, 4 numbers $(n + 1)$, right sides 1, 2, 3 (n) .

Question 6 PRF-12 · Mathematical proof · spec Prf2

For a positive integer n it is known that: (i) if n is a multiple of 4 then n is not a multiple of 6; (ii) n is a multiple of 12 or n is a multiple of 4. What can be deduced?

Answer: A

If n were a multiple of 12 it would be a multiple of both 4 and 6, contradicting (i). So by (ii) n is a multiple of 4, and by (i) not a multiple of 6.

Question 7 PRF-15 · Mathematical proof · spec Prf3

Let $u_n = n^2 - n + 41$. Which of the following statements is/are true?

- I u_n is prime for $n = 1, 2, 3, 4, 5$.
- II u_n is prime for all positive integers n .
- III u_{41} is divisible by 41.

Answer: F

$u_1 = 41, u_2 = 43, u_3 = 47, u_4 = 53, u_5 = 61$: all prime (I). $u_{41} = 41^2 - 41 + 41 = 41^2$, divisible by 41 (III) and not prime, so II is false - small cases do not prove a general claim.

Question 8 PRF-18 · Mathematical proof · spec Prf2

Every element of a set S of integers is either a multiple of 3 or a multiple of 5. Every multiple of 3 in S is negative. S contains the number 10. Which of the following statements must be true?

- I S contains a positive multiple of 5.
- II S contains no positive multiple of 3.
- III S contains a negative number.

Answer: E

$10 \in S$ is a positive multiple of 5 (I). Multiples of 3 in S are negative, so none is positive (II). III fails: $S = \{10\}$ satisfies everything.

Question 9 PRF-19 · Mathematical proof · spec Prf1

A student proves that $\sqrt{4}$ is irrational by copying the standard proof for $\sqrt{2}$: 'Suppose $\sqrt{4} = p/q$ in lowest terms; then $p^2 = 4q^2$, so p^2 is even, so p is even, $p = 2k$; then $4k^2 = 4q^2$, so $k^2 = q^2 \dots$ ' At which point does the argument break down compared with the $\sqrt{2}$ proof?

Answer: C

For $\sqrt{2}$ the step $p^2 = 2q^2$ leads to $2k^2 = q^2$, making q even. Here $k^2 = q^2$ carries no parity information, so the contradiction never appears - as expected since $\sqrt{4} = 2$.